SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN COURSE CODE: CSA1104

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4 **Total Marks:40**

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:
I, 192420006 B.Harshitha(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: B.Harshitha

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/ technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/ technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

Aim

To develop a UML model for a Book Bank System that verifies student details, obtains approval from the central computer, and facilitates the issue of required books through the Book Bank Office.

System Description

The Book Bank System manages the process of issuing books to eligible students. The administrator enters the student's details into the central computer. The central computer verifies the submitted student information and generates an approval status. After successful verification and approval, the office issues the books requested by the student.

The major activities are:

1. Student provides identification/details.
2. Administrator enters the student details into the central computer.
3. Central computer verifies the student details.
4. System generates approval.
5. Office checks the approval.
6. Student requests the required books.
7. Office issues the books to the student.
8. The transaction is recorded.

Actors

Actor	Responsibility
Student	Provides details and requests required books
Administrator	Enters student details into the central computer
Central Computer	Verifies student details and provides approval
Book Bank Office	Checks approval and issues books

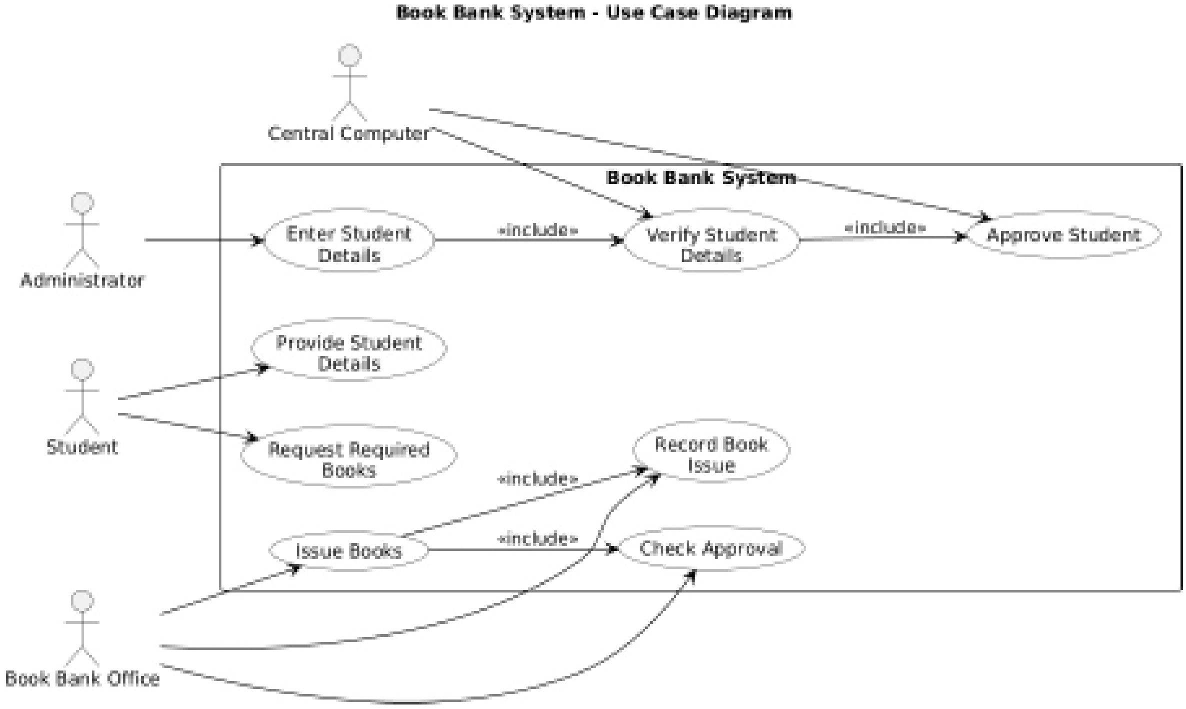
Main Use Cases

- Provide Student Details
- Enter Student Details
- Verify Student Details
- Approve Student
- Request Books
- Check Approval
- Issue Books
- Record Book Issue

Use Case Relationships

- Verify Student Details <<include>> Check Student Details
- Issue Books <<include>> Check Approval
- Issue Books <<include>> Record Book Issue
- Request Books occurs before Issue Books
- Approve Student follows successful verification

UML Use Case Diagram



Explanation of the UML Diagram

The Student provides the required details and requests the books. The Administrator enters the student's information into the system. The Central Computer verifies the submitted details and provides approval when the verification is successful.
The Book Bank Office checks the approval before issuing the requested books. After the books are issued, the transaction is recorded for maintaining the book bank records.
The <<include>> relationships represent activities that are essential parts of the corresponding process.

Analysis of the System

The UML model clearly identifies the interaction between the Student, Administrator, Central Computer, and Book Bank Office. The system separates student verification from book issuing, ensuring that books are issued only after the student's details have been verified and approved.
The use case model provides a systematic representation of the Book Bank process and helps identify the responsibilities of each actor. This improves the clarity, consistency, and maintainability of the system design.

Result

The Book Bank System was successfully modeled using UML. The use case diagram represents student verification, approval, book request, and book issue processes, along with the interactions between the Student, Administrator, Central Computer, and Book Bank Office.

Conclusion

The UML model provides a clear and structured representation of the Book Bank process. It ensures that student details are verified before approval and that books are issued only after the required verification and approval process is completed.

2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

Aim

To develop a UML model for an Exam Registration Portal that verifies candidate details, obtains approval from the central computer, and issues the hall ticket to the eligible candidate.

System Description

The Exam Registration Portal manages the candidate verification and hall-ticket issuing process. The administrator enters the candidate's details into the central computer. The central computer verifies the submitted information and provides approval if the candidate is eligible. After approval, the office issues the hall ticket to the candidate.

The major activities are:

1. Candidate provides registration details.
2. Administrator enters candidate details into the central computer.
3. Central computer verifies candidate details.
4. System provides approval.
5. Office checks the approval.
6. Hall ticket is generated and issued.
7. Candidate receives the hall ticket.

Actors

Actor	Responsibility
Candidate	Provides details and receives the hall ticket
Administrator	Enters candidate details into the system
Central Computer	Verifies candidate details and provides approval
Office	Checks approval and issues the hall ticket

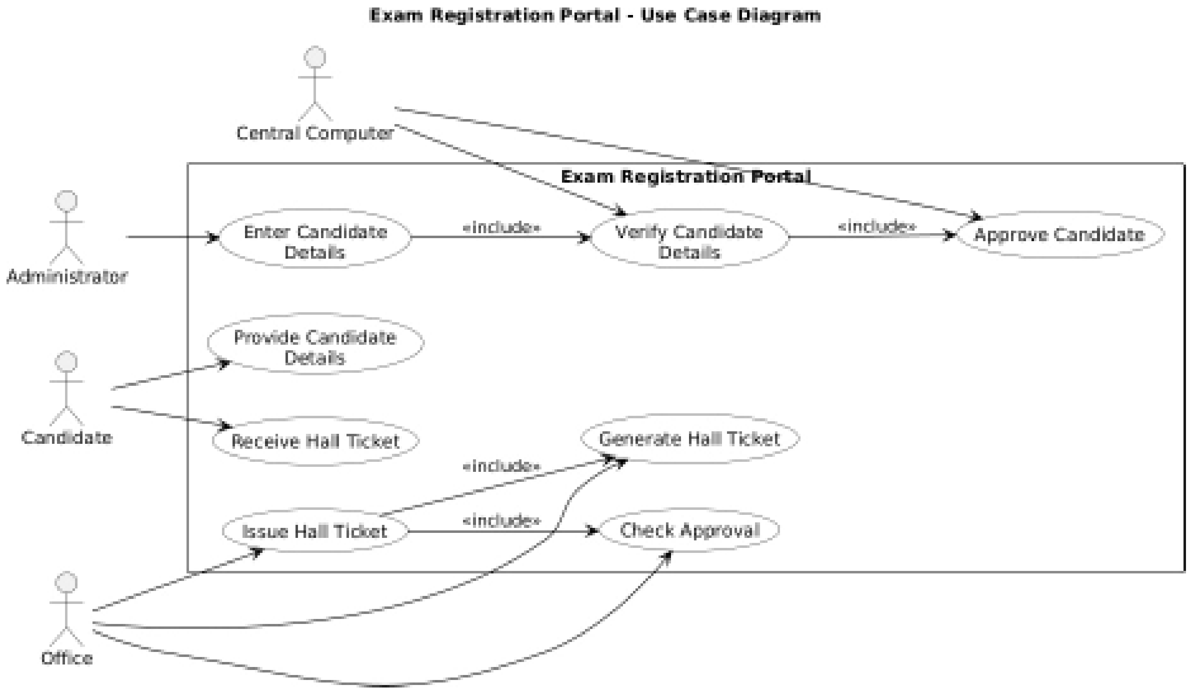
Main Use Cases

- Provide Candidate Details
- Enter Candidate Details
- Verify Candidate Details
- Approve Candidate
- Check Approval
- Generate Hall Ticket
- Issue Hall Ticket
- Receive Hall Ticket

Use Case Relationships

- Verify Candidate Details <<include>> Check Candidate Details
- Approve Candidate follows successful verification.
- Issue Hall Ticket <<include>> Check Approval
- Issue Hall Ticket <<include>> Generate Hall Ticket
- Candidate receives the hall ticket after successful approval.

UML Use Case Diagram



Explanation of the UML Diagram

The Candidate provides the required registration details and receives the hall ticket after successful verification.

The Administrator enters the candidate's details into the Exam Registration Portal.

The Central Computer verifies the candidate's details and provides approval.

The Office checks the approval, generates the hall ticket, and issues it to the candidate. The

<<include>> relationships represent activities that are essential components of the corresponding process.

Analysis of the System

The UML model clearly represents the interaction among the Candidate, Administrator, Central Computer, and Office. The candidate verification process is performed before the hall ticket is issued, ensuring that only an approved candidate receives the hall ticket.

The separation of candidate data entry, verification, approval, and hall-ticket issuing makes the system systematic and easy to understand.

Result

The Exam Registration Portal was successfully modeled using UML. The use case diagram represents candidate registration, verification, approval, and hall-ticket issuing processes with appropriate actor interactions.

Conclusion

The UML model provides a clear representation of the Exam Registration Portal. It ensures that candidate details are verified by the central computer and approval is obtained before the hall ticket is issued by the office. This provides a structured and reliable examination registration process.

3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

Aim

To develop a UML model for a Stock Maintenance System that manages customer orders, verifies orders, facilitates item purchases and delivery, and updates stock details after the transaction.

System Description

The Stock Maintenance System manages the ordering, verification, delivery, and stock updating process. The customer places an order with the assistance of the stock dealer. The central stock system verifies the order and checks the availability of the requested items. Once the order is verified, the items are delivered to the customer. After delivery, the stock details are updated in the central stock system.

The major activities are:

1. Customer selects the required items.
2. Customer places an order.
3. Stock dealer processes the order.
4. Central stock system verifies the order.
5. System checks item availability.
6. Order is confirmed.
7. Items are delivered to the customer.
8. Stock details are updated.

Actors

Actor	Responsibility
Customer	Places orders and purchases items
Stock Dealer	Assists the customer and processes orders
Central Stock System	Verifies orders, checks stock, and updates stock details

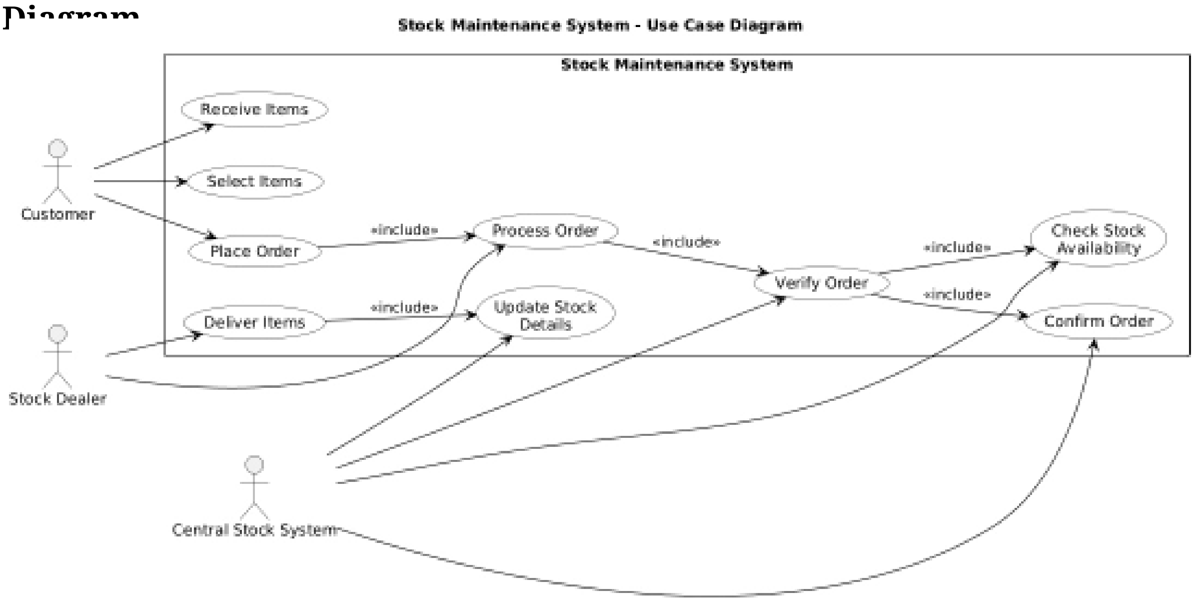
Main Use Cases

- Select Items
- Place Order
- Process Order
- Verify Order
- Check Stock Availability
- Confirm Order
- Deliver Items
- Update Stock Details
- Receive Items

Use Case Relationships

- Place Order <<include>> Process Order
- Process Order <<include>> Verify Order
- Verify Order <<include>> Check Stock Availability
- Deliver Items occurs after successful order verification.
- Update Stock Details occurs after items are delivered.
- Customer receives the delivered items.

UML Use Case Diagram



Explanation of the UML Diagram

The Customer selects the required items and places an order. The Stock Dealer assists the customer, processes the order, and facilitates delivery.

The Central Stock System verifies the order and checks the availability of the requested items. If the order is successfully verified, the order is confirmed and the items are delivered to the customer. After delivery, the stock system updates the stock details to reflect the items that have been purchased.

Analysis of the System

The UML model identifies the interaction between the Customer, Stock Dealer, and Central Stock System. The ordering process is separated into order placement, processing, verification, stock availability checking, delivery, and stock updating.

The model ensures that stock information is updated after items are delivered. This maintains accurate stock records and supports efficient order processing.

Result

The Stock Maintenance System was successfully modeled using UML. The use case diagram represents the complete process from placing and verifying an order to delivering items and updating the stock details.

Conclusion

The UML model provides a structured representation of the Stock Maintenance System. It ensures that customer orders are processed and verified before delivery, while the central stock system maintains updated stock information after each purchase.

4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Aim

To develop a UML model for an Online Course Reservation System that allows students to browse and select courses, checks seat availability through the university, and enrolls eligible students in available courses.

System Description

The Online Course Reservation System is managed by the central management system. The student first browses the available courses and selects a course of interest. The university then checks whether seats are available for the selected course. If a seat is available, the student is enrolled in the course.

The major activities are:

1. Student browses available courses.
2. Student selects a desired course.
3. University checks course availability.
4. System checks seat availability.
5. Course reservation is confirmed if a seat is available.
6. Student is enrolled in the selected course.
7. Course reservation details are updated.

Actors

Actor	Responsibility
Student	Browses courses, selects a course, and gets enrolled
University	Checks course and seat availability
Central Management System	Manages course reservation and enrollment

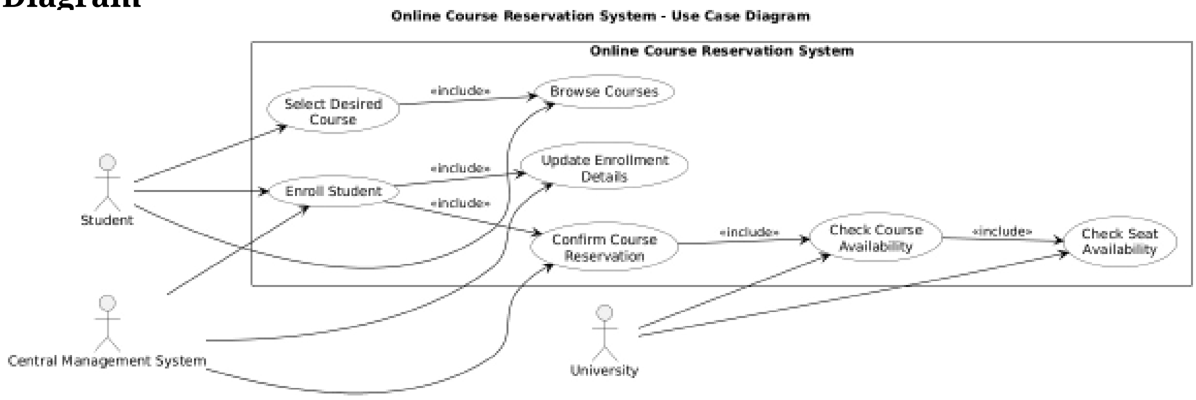
Main Use Cases

- Browse Courses
- Select Course
- Check Course Availability
- Check Seat Availability
- Confirm Reservation
- Enroll Student
- Update Enrollment Details

Use Case Relationships

- Select Course occurs after Browse Courses.
- Check Course Availability <<include>> Check Seat Availability.
- Confirm Reservation follows successful availability checking.
- Enroll Student occurs after reservation confirmation.
- Enroll Student <<include>> Update Enrollment Details.

UML Use Case Diagram



Explanation of the UML Diagram

The Student first browses the available courses and selects the desired course. The University checks the availability of the selected course and verifies whether seats are available.
The Central Management System manages the reservation and enrollment process. If a seat is available, the reservation is confirmed and the student is enrolled in the selected course. The enrollment details are then updated in the system.

Analysis of the System

The UML model clearly represents the interaction between the Student, University, and Central Management System. The process separates course browsing, course selection, availability checking, reservation confirmation, and student enrollment. The seat availability check acts as an important decision point. A student can be enrolled only when a seat is available. This provides an organized and systematic course reservation process.

Result

The Online Course Reservation System was successfully modeled using UML. The use case diagram represents course browsing, course selection, seat availability checking, reservation confirmation, and student enrollment.

Conclusion

The UML model provides a clear representation of the Online Course Reservation System. It ensures that students can select desired courses and that seat availability is checked before enrollment. The model supports systematic course reservation and enrollment management.